

DBMS

Normalize the following to 3NF

StudentNo	AdvisorNo	AdvisorName	AdvisorRoom	class1	Class2
11/02434	0129	Kenga	TC3-33	DBIT1	BIT3
12/02345	0128	Wanjiku	TC3-34	BIT4	DBIT1

1NF - Remove Repeating Groups:

StudentNo	AdvisorNo	AdvisorName	AdvisorRoom	class
11/02434	0129	Kenga	TC3-33	DBIT1
11/02434	0129	Kenga	TC3-33	BIT3
12/02345	0128	Wanjiku	TC3-34	BIT4
12/02345	0128	Wanjiku	TC3-34	DBIT1

2NF - Remove Partial Dependency:

StudentClass Table			Advisor Table		
StudentNo	Class	AdvisorNo	AdvisorNo	AdvisorName	AdvisorRoom
11/02434	DBIT1	0129	0129	Kenga	TC3-33
11/02434	BIT3	0129	0128	Wanjiku	TC3-34
12/02345	BIT4	0128			
12/02345	DBIT1	0128			

3NF - Remove Transitive Dependency:

StudentTable	Class Table	Advisor Table		
StudentNo	Class	AdvisorNo	AdvisorName	AdvisorRoom
11/02434	DBIT1	0129	Kenga	TC3-33
12/02345	BIT4	0128	Wanjiku	TC3-34
	BIT3			

StudentClass Table		
StudentNo	Class	AdvisorNo
11/02434	DBIT1	0129
11/02434	BIT3	0129
12/02345	BIT4	0128
12/02345	DBIT1	0128

OrderNo	CustomerNo	Cust Name	Cust Address	City	Date	Product Name	Desc	Quantity	Unit
1234	9876	Billy	Githurai	Nairobi	11/4/25	A123	Pencil	100	3.00
1234	9876	Billy	Githurai	Nairobi	11/4/25	B234	Eraser	200	1.50
1234	9876	Billy	Githurai	Nairobi	11/4/25	C345	Sharpener	5	8.00

1NF

Table is already in 1NF

2NF

Customer Table			
CustomerNo	Cust Name	Cust Address	City
9876	Billy	Githurai	Nairobi

Order Details Table						
OrderNo	CustomerNo	Date	Product Name	Desc	Quantity	Unit
1234	9876	11/4/25	A123	Pencil	100	3.00
1234	9876	11/4/25	B234	Eraser	200	1.50
1234	9876	11/4/25	C345	Sharpener	5	8.00

3NF

Customer Table			
CustomerNo	Cust Name	Cust Address	City
9876	Billy	Githurai	Nairobi

Order Details Table						
OrderNo	CustomerNo	Date	Product Name	Desc	Quantity	Unit
1234	9876	11/4/25	A123	Pencil	100	3.00
1234	9876	11/4/25	B234	Eraser	200	1.50
1234	9876	11/4/25	C345	Sharpener	5	8.00